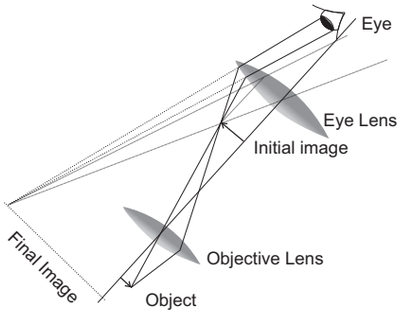
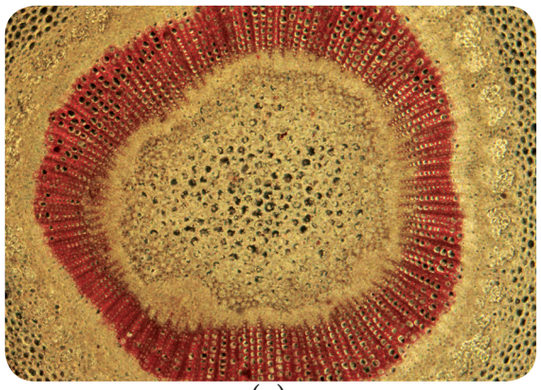






(a) (b) (c)

Figure 6.2: a. Light microscope; **b.** Ray diagram - light path; c. Image taken using light microscope;

There are two kinds of electron microscopes namely:

1. Transmission Electron Microscope (TEM)
2. Scanning Electron Microscope (SEM)

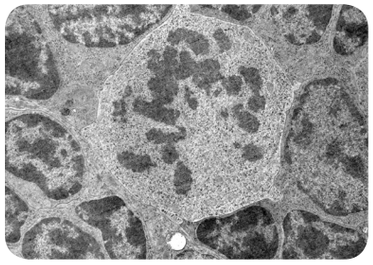
1. Transmission electron microscope:

This is the most commonly used electron microscope which provides two dimensional image. The components of the microscope are as follows:

a. Electron generating system
b. Electron condensor
c. Specimen objective
d. Tube lens
e. Projector

A beam of electron passes through the specimen to form an image on fluorescent screen. The magnification is 1–3 lakhs times and resolving power is 2–10 Å. It is used for studying detailed structrue of viruses, mycoplasma, cellular organelles, etc (Figure 6.3 a and b).





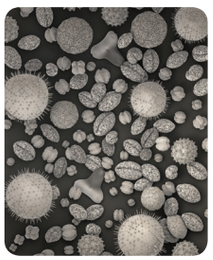
(a) (b)

Figure 6.3: a. Transmission electron microscope; b. Image of TEM

2. Scanning Electron Microscope:

This is used to obtain three dimensional image and has a lower resolving power than TEM. In this, electrons are focused by means of lenses into a very fine point. The interaction of electrons with the specimen results in the release of different forms of radiation (such as auger electrons, secondary electrons, back scattered electrons) from the surface of the specimen. These radiations are then captured by an appropriate detector, amplified and then imaged on fluorescent screen. The magnification is 2,00,000 times and resolution is 5–20 nm (Figure 6.4 a and b).





(a) (b)

Figure 6.4: a. Scanning electron microscope; b. Image of SEM

6.3. Cell Theory

In 1833, German botanist **Matthias Schleiden** and German zoologist **Theodor Schwann** proposed that all plants and animals are composed of cells and that cells were the basic building blocks of life.

XIth_Botany_Chapter_6_EM.indd 146 5/17/2020 9:09:43 PM